

Updated Project Vision

Project Name

MedExplain AI – AI Healthcare Assistant Platform

Project Goal

Build a production-ready AI Healthcare Assistant Platform using Django REST Framework, PostgreSQL, JWT Authentication, and Google Gemini 2.5 Pro.

The application should help non-medical users easily understand their medical reports in their preferred language.

The platform should not only analyze medical reports but also provide an intelligent healthcare assistant experience.

The application must be modular, scalable, secure, and production-ready.

Core Features

1. User Authentication

- Register
 - Login
 - JWT Authentication
 - Refresh Token
 - Logout
 - Forgot Password
 - Change Password
 - User Profile
-

2. Medical Report Upload

Users can upload

- PDF
- PNG
- JPG

- JPEG

Supported reports include

- CBC
 - Blood Test
 - Urine Test
 - Thyroid Profile
 - Liver Function Test (LFT)
 - Kidney Function Test (KFT)
 - Lipid Profile
 - HbA1c
 - Vitamin Reports
 - Any pathology report
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3. AI Report Analysis

The AI should

Extract

- Patient Name
- Age
- Gender
- Doctor Name
- Hospital / Clinic / Laboratory
- Address
- Phone Number
- Report Date

Extract every laboratory test

Example

- Hemoglobin
- RBC
- WBC
- Platelets
- Glucose
- Cholesterol
- HDL
- LDL
- Triglycerides
- Creatinine
- Bilirubin
- SGOT

- SGPT
- TSH
- HbA1c

Extract

- Value
- Unit
- Reference Range

Return everything as structured JSON.

Never return unstructured text.

4. Medical Rule Engine

Do not depend on AI to determine whether a result is normal.

The backend should compare every value against the reference range.

Generate

- Normal
- Low
- High
- Critical (where applicable)

Calculate an overall report status

- Normal
 - Needs Attention
 - Critical
-

5. Possible Associated Conditions

The system should never claim that the patient has a confirmed disease.

Instead generate

Possible Associated Conditions

Example

Low Platelets

Possible Associated Conditions

- Dengue
- Viral Infection
- Immune Thrombocytopenia (ITP)

Always mention that only a qualified healthcare professional can diagnose a medical condition.

6. AI Healthcare Assistant Chat

After report analysis, the user can ask questions.

Examples

"What is Hemoglobin?"

"Why are my Platelets low?"

"What foods may help improve Hemoglobin?"

"What does high cholesterol mean?"

"Explain my report like I'm 10 years old."

"Can you explain this report in Gujarati?"

The chatbot should answer using the uploaded report as context.

7. Multilingual Support

The user can select

- English
- Hindi
- Gujarati
- Marathi
- Tamil
- Telugu
- Bengali

All explanations, summaries, and chatbot responses should be generated in the selected language.

8. Previous Report Comparison

Store every uploaded report.

Allow the user to compare

Current Report

vs

Previous Reports

Display trends for

- Hemoglobin
- WBC
- RBC
- Platelets
- Blood Sugar
- Cholesterol
- HbA1c
- Creatinine
- TSH

Generate charts showing improvement or decline over time.

9. Health Trend Analysis

The system should automatically detect

Increasing values

Decreasing values

Sudden abnormal changes

Example

Hemoglobin

January 11.5

March 10.2

June 8.7

Trend

Declining

Recommendation

Discuss this trend with your healthcare provider.

10. Health Alerts

Generate alerts when

Critical values are detected.

Values change significantly compared with previous reports.

Example

Platelets decreased by 55% since the previous report.

Blood sugar has remained above the normal range for three consecutive reports.

Hemoglobin has steadily decreased over the last six months.

11. AI Summary

Generate a simple health summary.

Example

Your blood sugar is above the normal range.

Your platelet count is lower than expected.

Your hemoglobin level is low.

Some findings may require medical evaluation.

Please consult your healthcare professional.

12. Download Report

Allow users to download

AI Analysis PDF

The PDF should include

Patient Information

Test Results

Charts

Health Summary

Possible Associated Conditions

Recommendations

Date

Language

13. Dashboard

Dashboard should display

Profile

Upload Report

Report History

Recent Reports

Trend Charts

Health Alerts

AI Chat

Settings

14. AI Architecture

The project should follow this workflow

User Uploads PDF/Image



PDF Parser / OCR



Gemini 2.5 Pro



Structured JSON Extraction



Python Medical Rule Engine



Health Trend Analysis



Possible Associated Conditions



AI Explanation



Translation



Dashboard

15. Future Enhancements (Architecture Ready)

Design the project so these features can be added later without major changes

- RAG using trusted medical references
- Medical knowledge base
- Wearable device integration (Google Fit / Apple Health)
- Doctor portal
- Appointment booking
- Prescription management
- Medication reminders
- Email and WhatsApp notifications
- Cloud storage (AWS S3)
- Mobile application support

Important AI Rules

The AI must never claim a confirmed diagnosis.

The AI must explain laboratory findings in simple language.

The AI must generate structured JSON.

The AI must provide multilingual responses.

The AI must clearly state that the information is educational and not a substitute for professional medical advice.

Whenever medical conditions are mentioned, they should be presented as possible associated conditions, not confirmed diagnoses.

The AI should always be transparent about uncertainty and avoid making unsupported medical claims.